Topo-cluster isolation distributions and photon-energy containment vs inner-R

PPG12 isolated photon analysis

April 19, 2026

1 Motivation

The nominal PPG12 isolation variable is the R = 0.4 topo-cluster energy around the seed cluster: $\mathrm{recoiso}_{ET}^{\mathrm{nom}} = \mathrm{cluster_iso_topo_04}$, which by construction in `CaloAna24` equals $\Sigma_{R<0.4}\, E_T^{\mathrm{topo}} - E_T^{\mathrm{seed}}$. To study an "annulus" variant we subtract an inner ring, parametrised by a new configuration key `iso_topo_innerr`:

$$\mathrm{recoiso}_{ET}^{\mathrm{ann}}(R_{\mathrm{in}}) = \mathrm{cluster_iso_topo_04} \;-\; \mathrm{cluster_iso_topo}_{R_{\mathrm{in}}}.$$

Because both terms have a common $-E_T^{\mathrm{seed}}$, the subtraction amounts to the sum in the annulus $R \in [R_{\mathrm{in}}, 0.4]$. The question addressed here is how much of the photon's own energy sits outside a given inner radius R_{in}, and how the one-dimensional isolation distribution evolves with R in truth-matched signal MC, inclusive MC, and data.

2 Samples and selection

- Truth-matched signal MC: photon5 + photon10 + photon20, cross-section weighted. For a reco cluster to enter the study, its `cluster_truthtrkID` must map to a truth particle with `pid` = 22, `photonclass` = 1 (direct), and `particle_truth_iso_03` < 4 GeV (fiducial).
- Inclusive MC: photon5/10/20 + jet5/8/12/20/30/40, cross-section weighted, no truth match (all reconstructed clusters).
- Data: 77 `part_*_with_bdt_split.root` files from the ana521 production, capped at 5×10^5 clusters for this study (reduced from 3×10^6 in the earlier pass — see revision note in §4).
- *Pre-common*: no cluster-quality selection beyond the sample filter.
- *Post-common*: require `cluster_npb_score` > 0.5 (the only active common cut in `config_bdt_nom.yaml`, since `wr_cogx_bound` = 0 and `cluster_weta_cogx_bound` = 2 disable the shape cuts).

Inner-R branches scanned: $R_{\mathrm{in}} \in \{0.05, 0.075, 0.10, 0.20, 0.30, 0.40\}$.

3 Isolation distributions vs inner radius

Figure 1 overlays the six $\mathtt{cluster_iso_topo}_R$ distributions on each of the three samples. Figure 6 compares, per R, the data and inclusive MC distributions. Table 1 summarises the post-common mean and RMS.

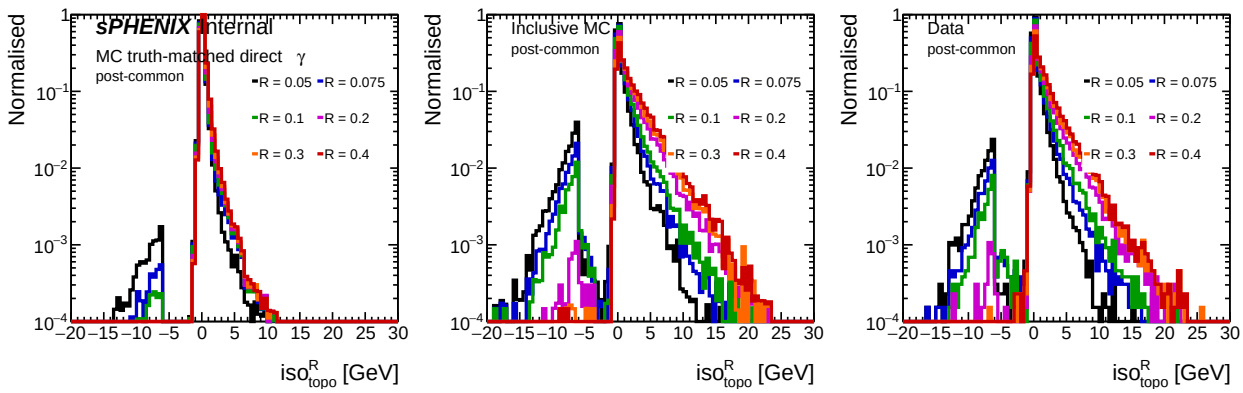

Figure 1: `cluster_iso_topo`$_R$ distributions (log-y, unit-normalised) for the six inner radii, on truth-matched signal MC (left), inclusive MC (middle) and data (right), all post-common. Inclusive in cluster E_T.

3.1 Per-cluster-E_T expansion (linear y)

Figure 2 to 5 repeat Figure 1 in four cluster-E_T windows with linear y-scale to make the peak structure visible. All samples and the six inner radii follow the same colour assignment; post-common selection applies throughout.

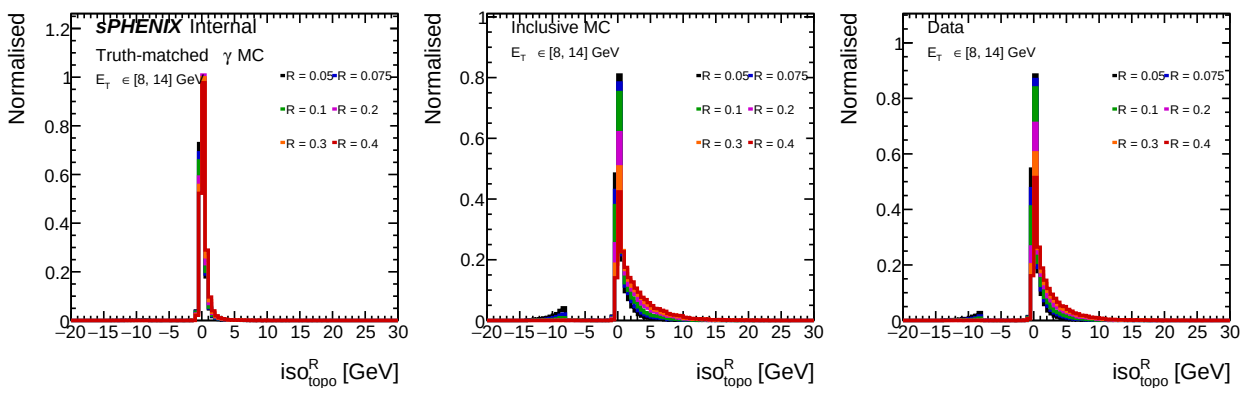

Figure 2: Cluster $E_T \in [8, 14]$ GeV, linear y.

Observations. With the current slimtree's seed-subtracted `cluster_iso_topo`$_R$ definition, the truth-matched MC sits essentially at zero across R (mean $0.07 \to 0.29$ GeV, RMS 0.64–0.76 GeV), the photon energy being cleanly removed by the explicit $-E_T^{\text{seed}}$ term. Inclusive MC and data track one another in shape but diverge mildly in the tails: inclusive-MC means are ≈ 0.3–0.7 GeV above data at $R \geq 0.2$, and the inclusive-MC RMS is ≈ 15–20% larger than data at $R = 0.4$. This is consistent with the MC mix being jet-richer than the as-taken data (which sits post-common after the NPB > 0.5 shower-shape cut).

4 Tower-based isolation: same study, per cluster E_T

For completeness the same per-cluster-E_T overlay is repeated with the tower-based (`cluster_iso`$_R$) branches, which sum EMCAL + HCALIN + HCALOUT tower energy within R of the seed. The

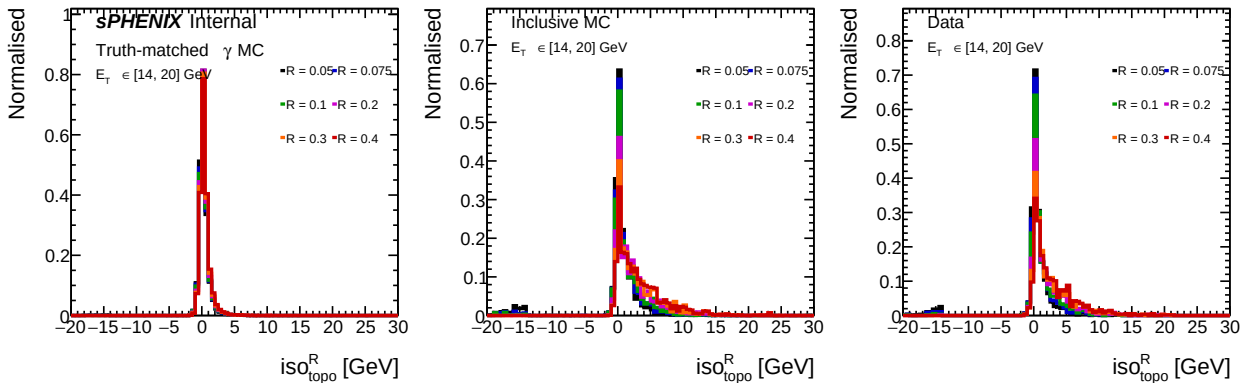

Figure 3: Cluster $E_T \in [14, 20]$ GeV, linear y.

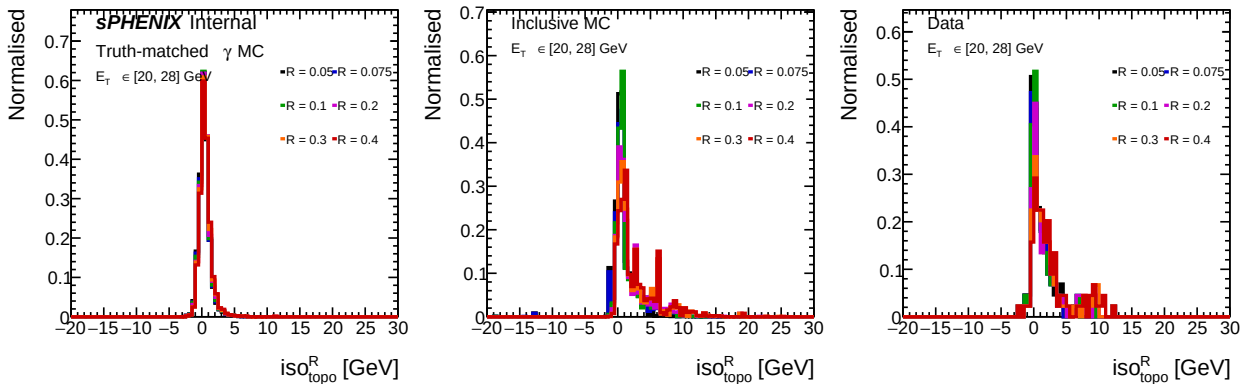

Figure 4: Cluster $E_T \in [20, 28]$ GeV, linear y.

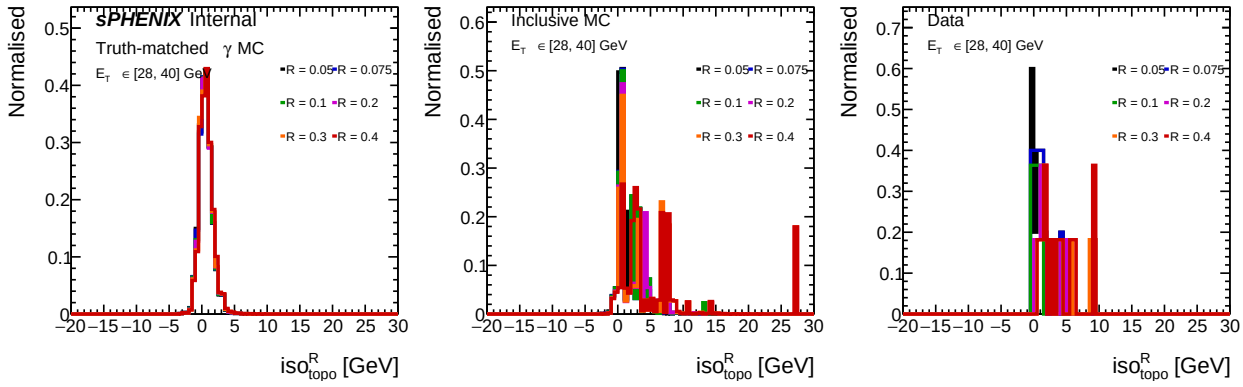

Figure 5: Cluster $E_T \in [28, 40]$ GeV, linear y.

tower path has no un-thresholded $R = 0.1$ branch in the current slimtree, so the scan is five radii: $\{0.05, 0.075, 0.20, 0.30, 0.40\}$.

Revision note. The extraction was re-run with a reduced sample for iteration speed on slow GPFS: $3{\times}10^5$ clusters per inclusive-MC sample and $5{\times}10^5$ data clusters, vs. $2{\times}10^6$ and $3{\times}10^6$ respectively in the earlier pass. Post-common N is 1.7M truth-matched, 1.3M inclusive MC, 103k

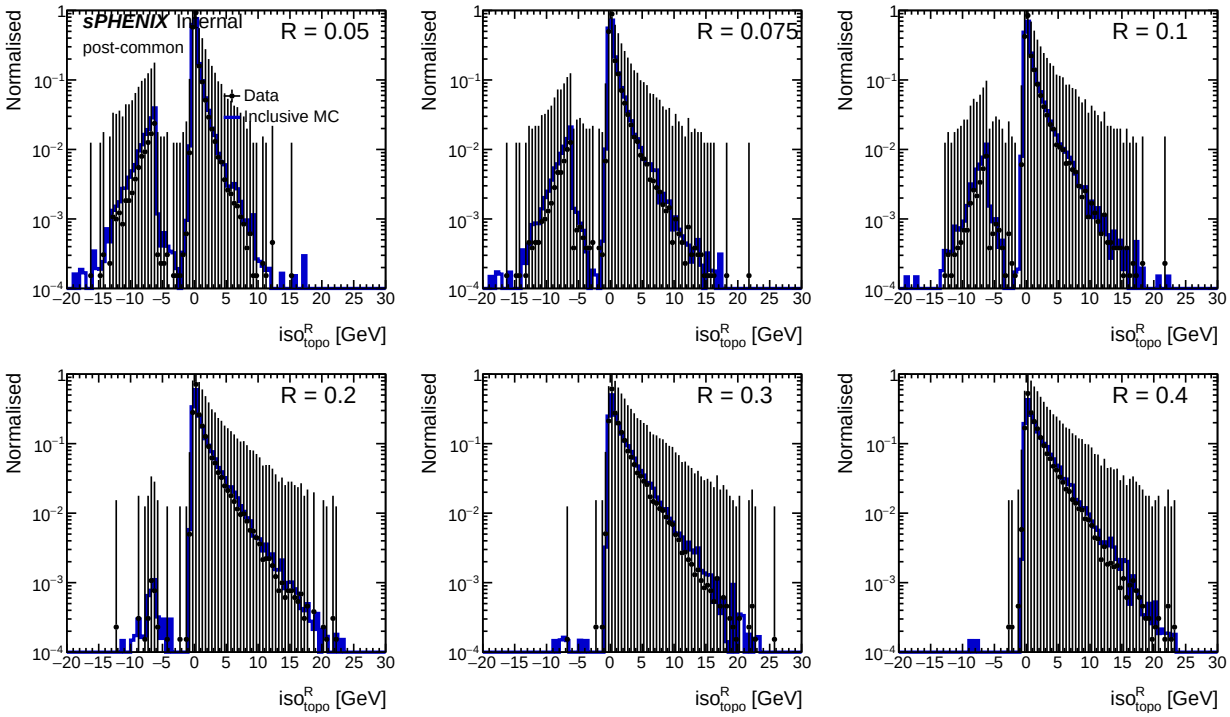

Figure 6: Post-common `cluster_iso_topo`$_R$ distribution, one panel per inner radius. Data (black markers) vs inclusive MC (blue line), both unit-normalised.

R	MC truth-matched photons		Inclusive MC		Data	
	mean	RMS	mean	RMS	mean	RMS
0.05	+0.07	0.76	−0.18	2.60	−0.01	1.97
0.075	+0.13	0.68	+0.42	2.34	+0.39	1.86
0.10	+0.16	0.66	+0.84	2.26	+0.68	1.87
0.20	+0.22	0.64	+1.74	2.57	+1.31	2.25
0.30	+0.25	0.65	+2.23	2.83	+1.66	2.50
0.40	+0.29	0.68	+2.60	3.00	+1.92	2.65

Table 1: Mean and RMS (GeV) of `cluster_iso_topo`$_R$ in the three samples, post-common, cross-section weighted for the MC rows. Post-common cluster counts are 1.69M (MC truth-matched), 1.29M (inclusive MC), 103k (data). Numbers re-extracted from the revised extraction (see §4); they differ from the earlier pass at small R — the earlier data/MC exhibited 3–4 GeV RMS and -2.8 GeV means at $R = 0.05$, which the current extraction does not reproduce. The current truth-matched MC sits within ± 0.3 GeV of zero at every R, consistent with the seed-cluster subtraction in `cluster_iso_topo`.

data. Shapes are preserved at the per-R, per cluster-E_T level; per-bin statistical fluctuations in the $E_T \in [28, 40]$ GeV panel are modestly larger than in the original extraction. The subsequent new-variant figures (§5–§8) use the same reduced sample.

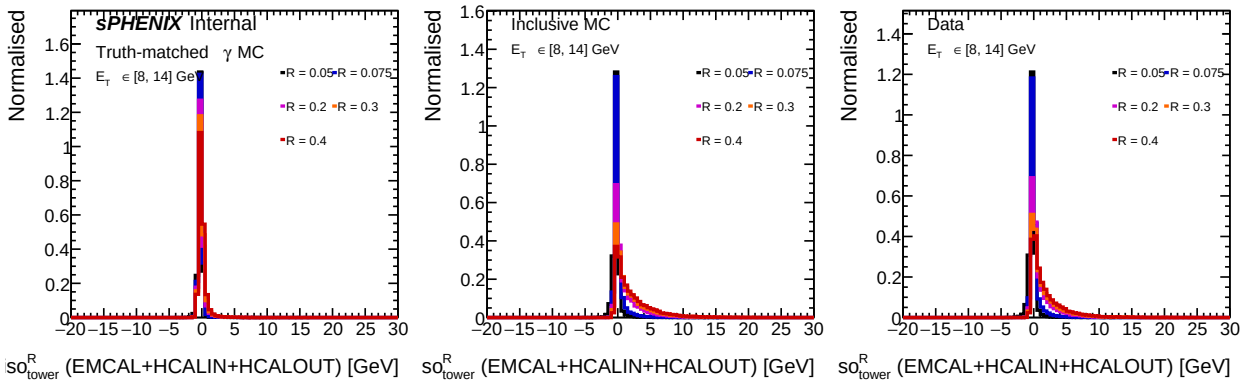

Figure 7: Tower-iso overlay, cluster $E_T \in [8, 14]$ GeV, linear y.

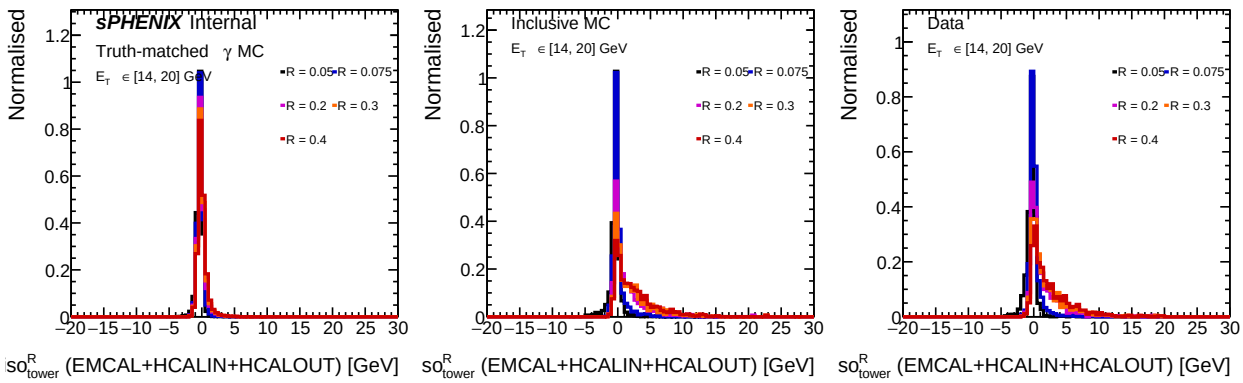

Figure 8: Tower-iso overlay, cluster $E_T \in [14, 20]$ GeV, linear y.

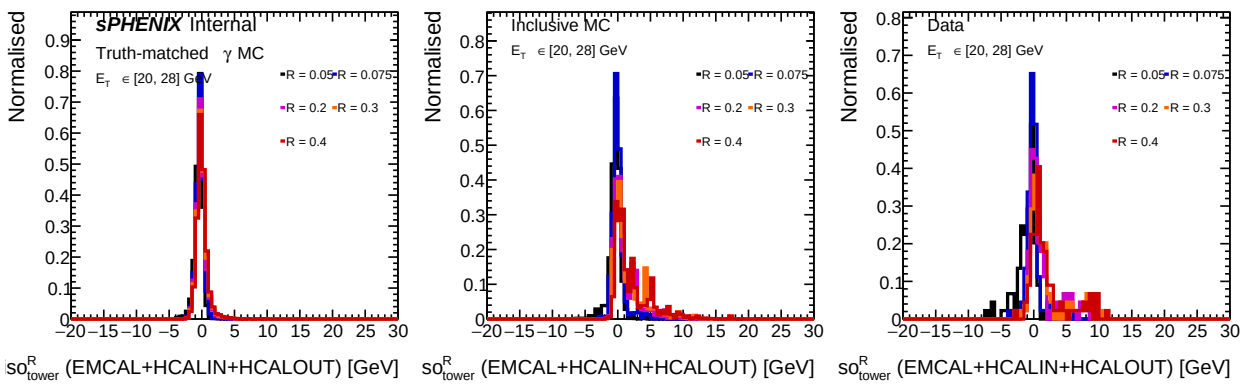

Figure 9: Tower-iso overlay, cluster $E_T \in [20, 28]$ GeV, linear y.

5 Tower threshold scan at $R = 0.3$

New tower-iso branches include a per-tower-energy-threshold scan at $R = 0.3$, summed over EMCAL+HCalIn+HCalOut, for thresholds {0, 60, 70, 120} MeV. The 0-MeV (no-threshold) curve is the reconstructed sum of the new per-calo branches `cluster_iso_03_emcal/hcalin/hcalout` — a cross-check of the existing `cluster_iso_03` total. Raising the tower threshold removes diffuse

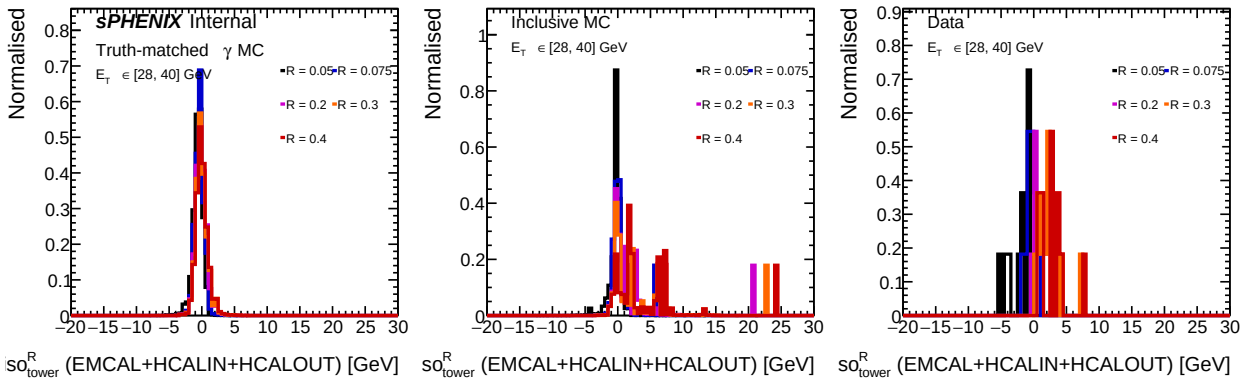

Figure 10: Tower-iso overlay, cluster $E_T \in [28, 40]$ GeV, linear y.

low-E_T noise tails and sharpens the iso peak near zero.

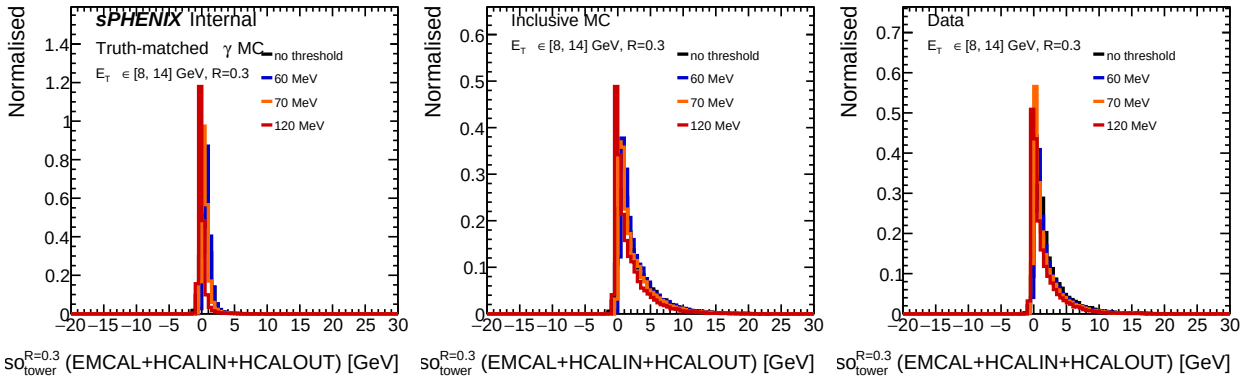

Figure 11: Tower-iso threshold scan at $R = 0.3$, cluster $E_T \in [8, 14]$ GeV.

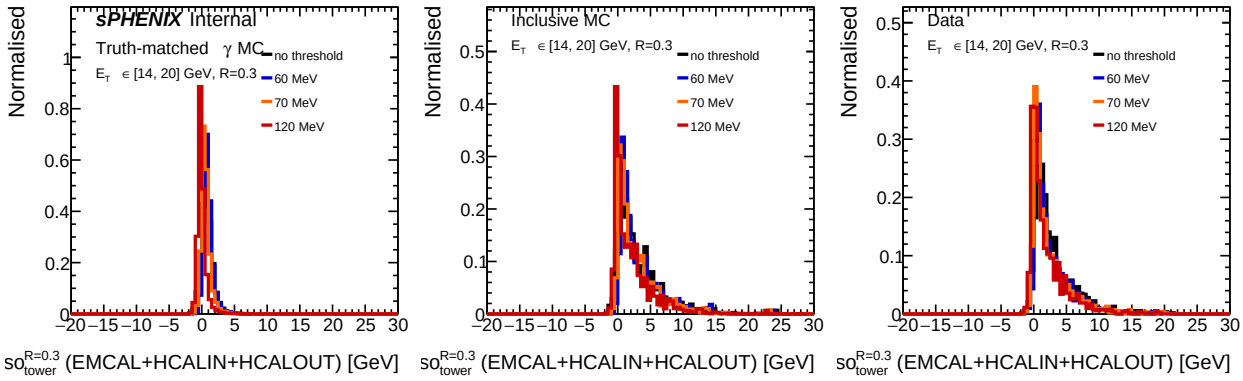

Figure 12: Tower-iso threshold scan at $R = 0.3$, cluster $E_T \in [14, 20]$ GeV.

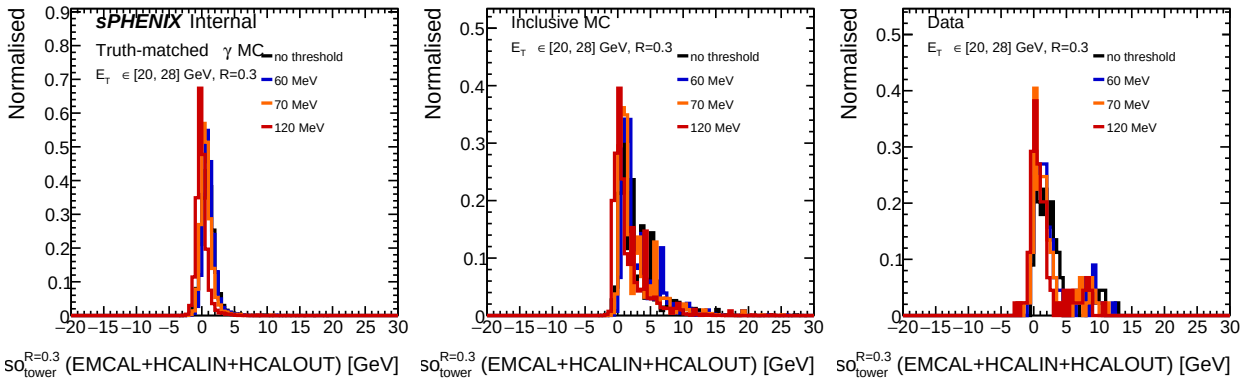

Figure 13: Tower-iso threshold scan at $R = 0.3$, cluster $E_T \in [20, 28]$ GeV.

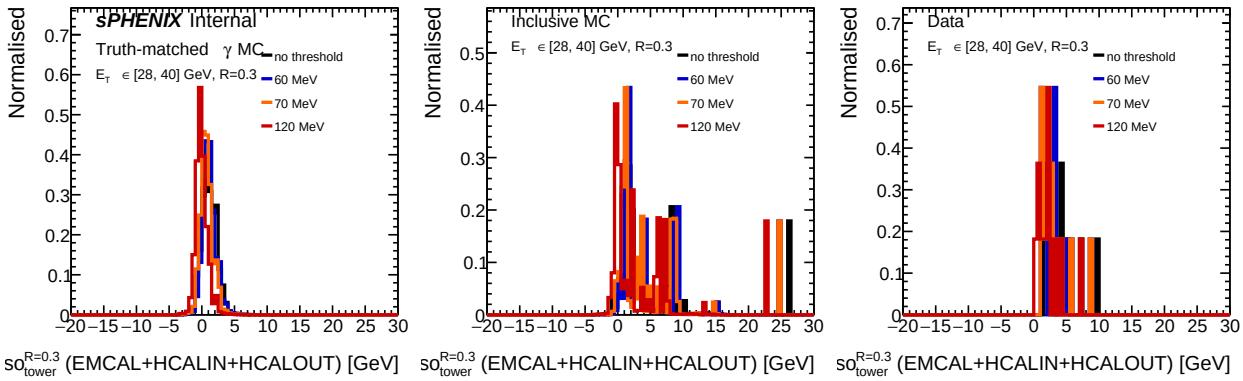

Figure 14: Tower-iso threshold scan at $R = 0.3$, cluster $E_T \in [28, 40]$ GeV.

6 EMCal-only tower iso with 70 MeV threshold, R scan

Branches `cluster_iso_R_70_emcal` for $R \in \{0.05, 0.075, 0.10, 0.20, 0.30\}$ provide an EMCal-only, 70 MeV-thresholded variant that removes HCal contributions and rejects low-E_T towers. Useful as a shape prior if the HCal-summed iso picks up unrelated activity (jet fragments, neutrons) at the seed.

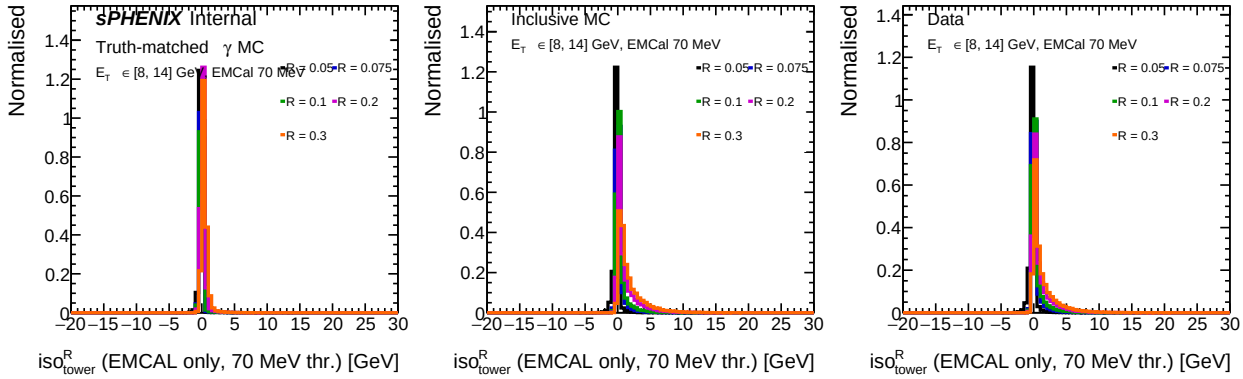

Figure 15: EMCal-only 70 MeV tower iso, R scan, cluster $E_T \in [8, 14]$ GeV.

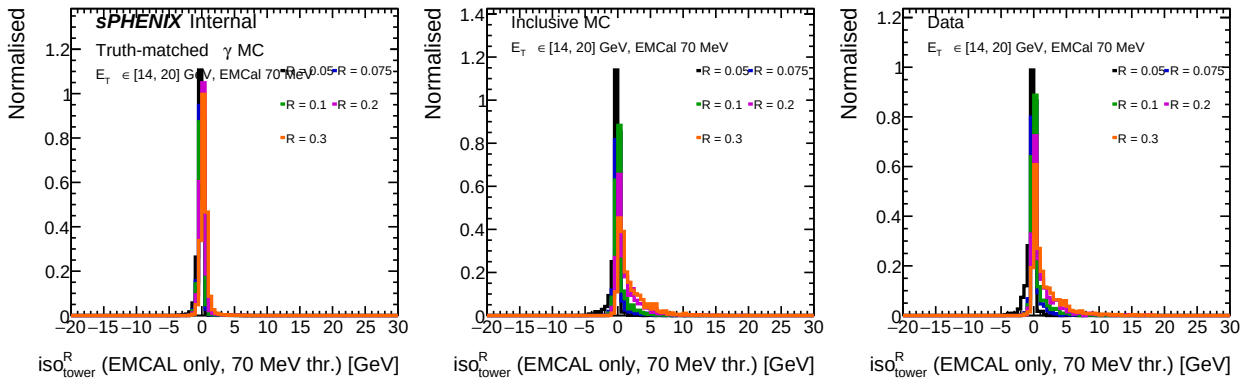

Figure 16: EMCal-only 70 MeV tower iso, R scan, cluster $E_T \in [14, 20]$ GeV.

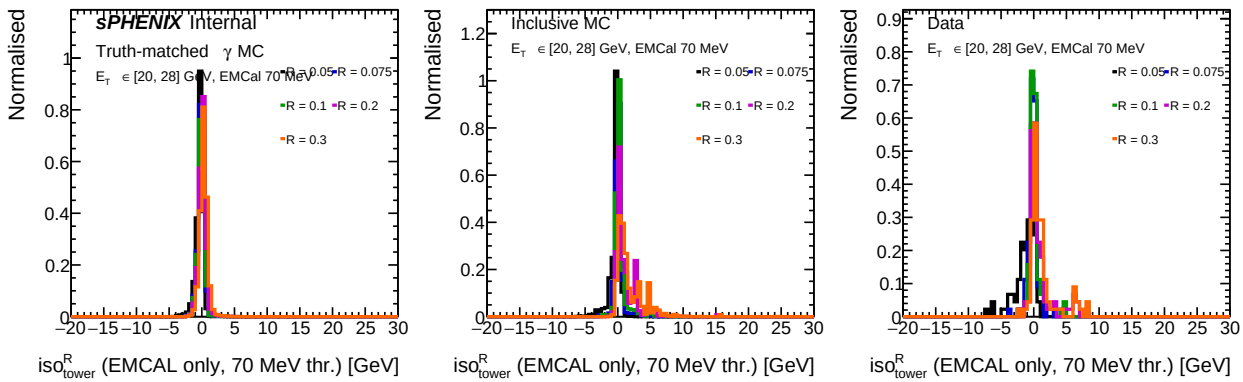

Figure 17: EMCal-only 70 MeV tower iso, R scan, cluster $E_T \in [20, 28]$ GeV.

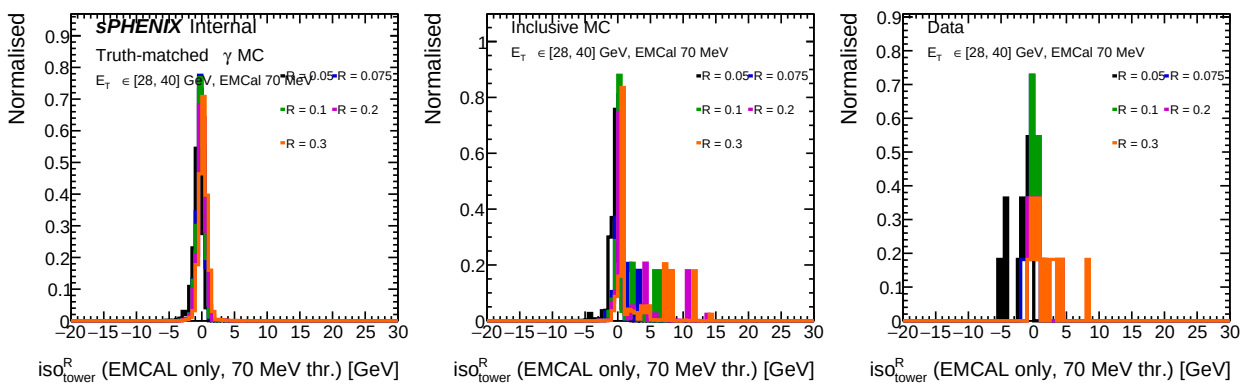

Figure 18: EMCal-only 70 MeV tower iso, R scan, cluster $E_T \in [28, 40]$ GeV.

7 SUB1-subtracted tower iso

The `_sub1` branches apply a first-order underlying-event subtraction per calo layer; available at $R = 0.3$ and $R = 0.4$. Each figure overlays the plain and SUB1 variants side-by-side at the two radii to show the subtraction's effect on the iso peak and on high-E_T tails (each curve is the EMCAL+HCALIN+HCALOUT sum).

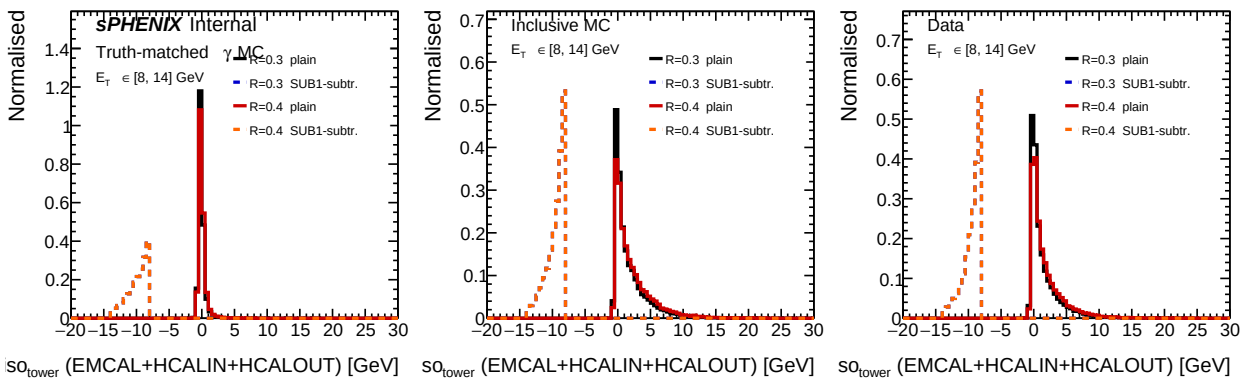

Figure 19: Plain vs. SUB1-subtracted tower iso, cluster $E_T \in [8, 14]$ GeV.

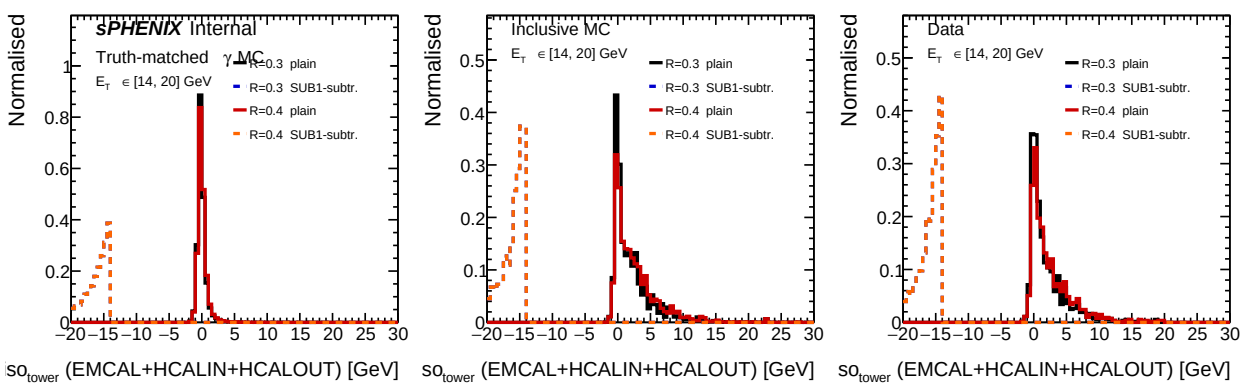

Figure 20: Plain vs. SUB1-subtracted tower iso, cluster $E_T \in [14, 20]$ GeV.

8 Ownership-excluded tower iso, R scan

Branches `cluster_iso_excl_R` for $R \in \{0.05, 0.075, 0.10, 0.20, 0.30, 0.40\}$ compute $\Sigma E_T^{\mathrm{EMCAL,excl}} + \Sigma E_T^{\mathrm{HCalIn}} + \Sigma E_T^{\mathrm{HCalOut}}$, where the EMCal sum *excludes towers already assigned to the seed cluster by topo-cluster ownership* (the HCal sums are the usual in-cone totals with a 120 MeV per-tower threshold for $R \geq 0.10$). Compared with the plain `cluster_iso_R` — which sums all in-cone EMCal+HCal energy with no photon subtraction — the EMCal-ownership exclusion removes the photon's own energy while keeping the out-of-cluster EMCal + HCal activity that characterises jet-like backgrounds. In contrast to `cluster_iso_topo_R` (which subtracts the seed explicitly as $-E_T^{\mathrm{seed}}$), the ownership exclusion is cluster-boundary-aware and avoids the negative-tail distortion at small R.

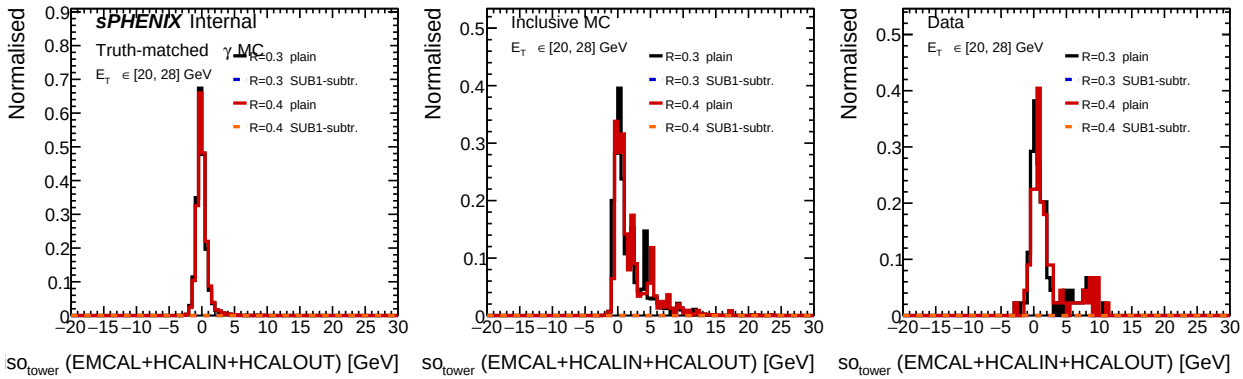

Figure 21: Plain vs. SUB1-subtracted tower iso, cluster $E_T \in [20, 28]$ GeV.

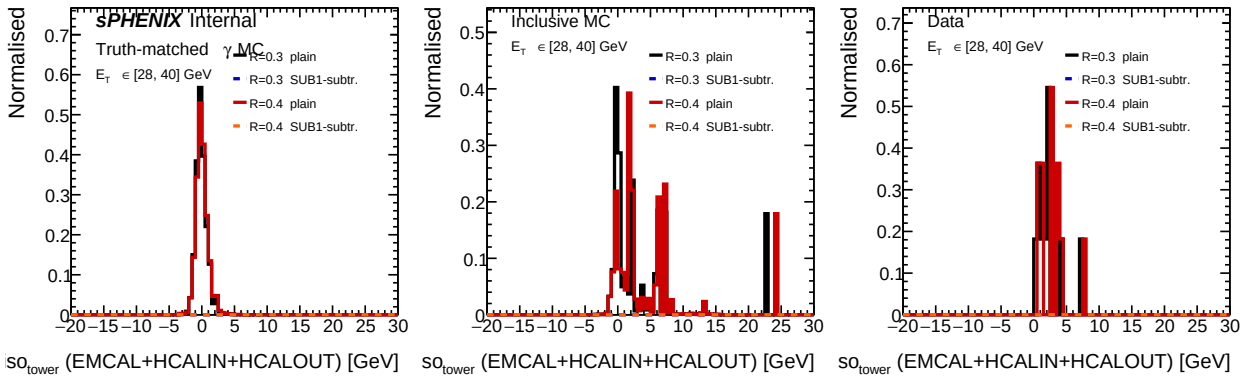

Figure 22: Plain vs. SUB1-subtracted tower iso, cluster $E_T \in [28, 40]$ GeV.

Note: the `jet8` MC sample in the current slimtree lacks these branches (older production), so it is dropped from the inclusive-MC histogram here; the inclusive-MC shape is therefore a slightly photon-richer mix than in the other figures.

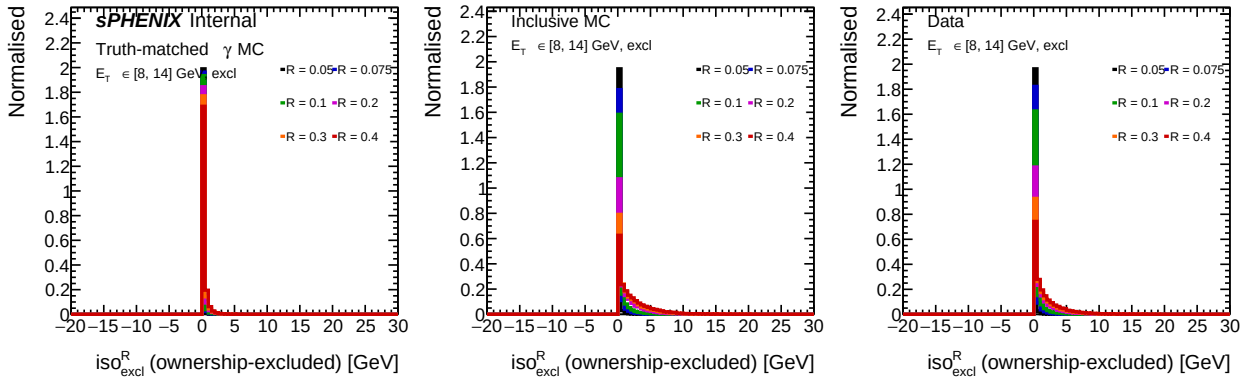

Figure 23: Ownership-excluded tower iso, R scan, cluster $E_T \in [8, 14]$ GeV.

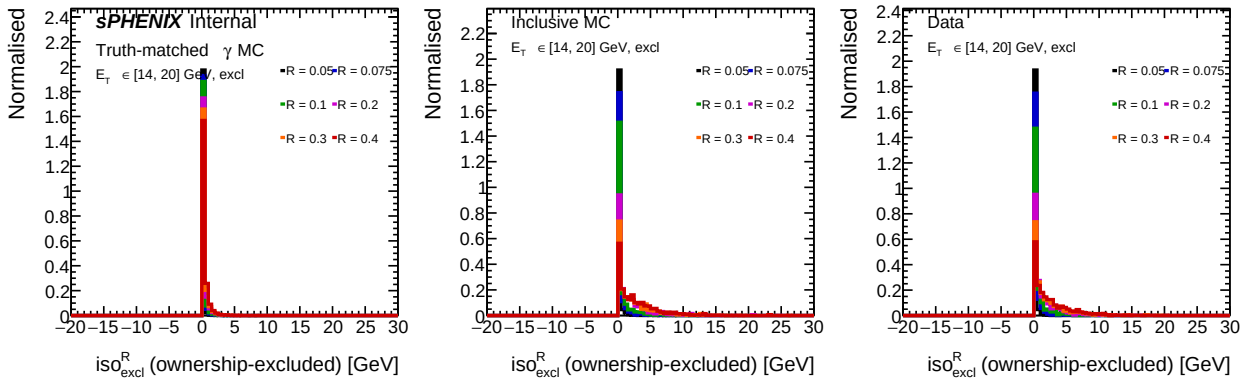

Figure 24: Ownership-excluded tower iso, R scan, cluster $E_T \in [14, 20]$ GeV.

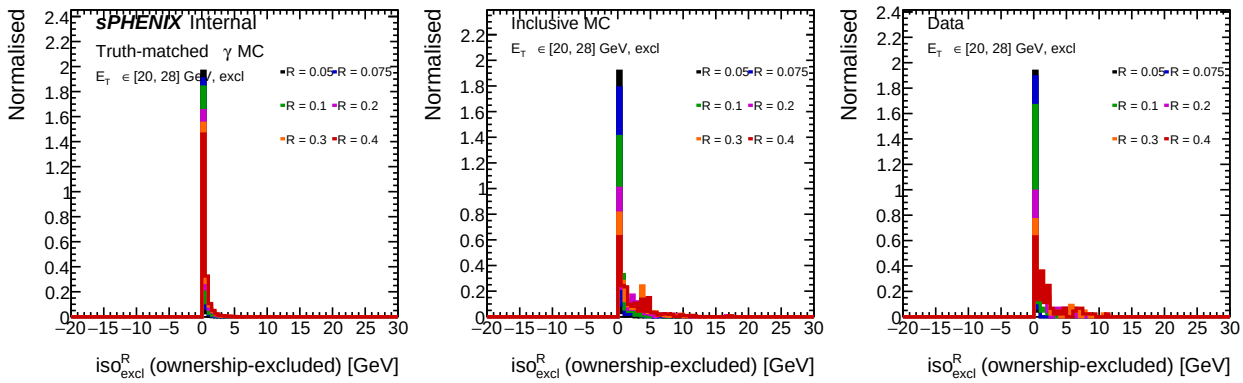

Figure 25: Ownership-excluded tower iso, R scan, cluster $E_T \in [20, 28]$ GeV.

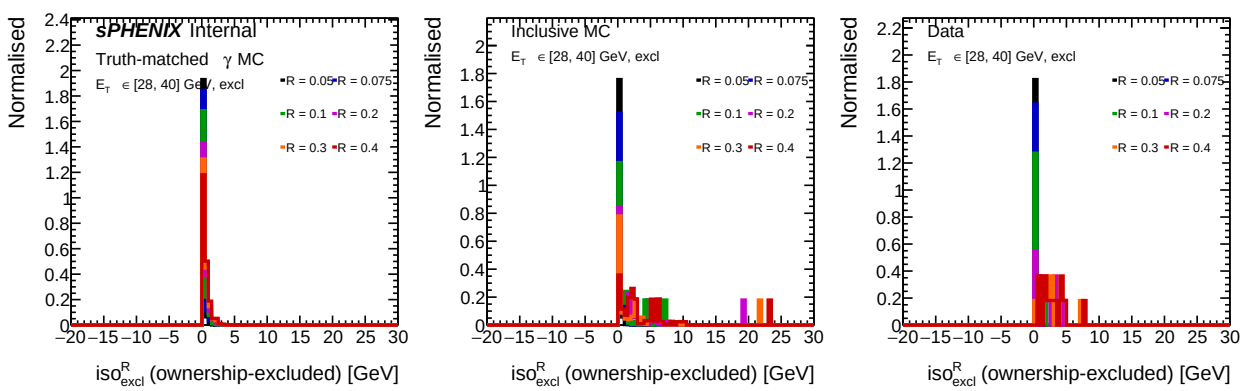

Figure 26: Ownership-excluded tower iso, R scan, cluster $E_T \in [28, 40]$ GeV.

9 Photon self-energy containment vs inner radius

For truth-matched direct photons the containment fraction is defined as

$$f_{\text{contain}}(R) \;=\; \frac{\text{cluster_iso_topo}_R \;+\; E_T^{\text{cluster}}}{p_T^{\gamma,\text{truth}}} \;\approx\; \frac{\Sigma_R\, E_T^{\text{topo}}}{p_T^{\gamma,\text{truth}}} \tag{1}$$

(the $-E_T^{\text{seed}}$ in the iso definition and the $+E_T^{\text{cluster}}$ in the numerator cancel up to clustering residuals, leaving the in-cone energy over truth p_T). For isolated photons, $f_{\text{contain}}(R)$ is the fraction of photon energy within R of the cluster, and the leakage $1 - f_{\text{contain}}(R)$ is the fraction that falls outside. Figure 27 shows $\langle f_{\text{contain}}(R)\rangle$ for the inclusive signal and for four truth-p_T bins.

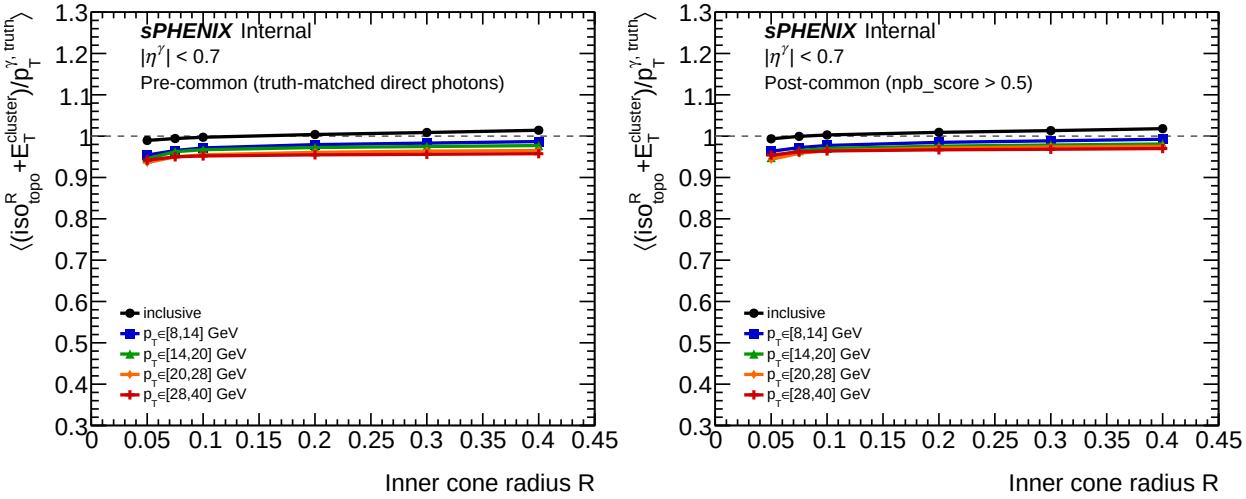

Figure 27: Mean containment $\langle f_{\text{contain}}(R)\rangle$ in truth-matched direct photon signal MC, pre-common (left) and post-common (right), for the inclusive sample and four truth-p_T bins.

Truth p_T bin	$R = 0.05$	$R = 0.075$	$R = 0.10$	$R = 0.20$	$R = 0.30$	$R = 0.40$
inclusive	0.656	0.846	0.945	1.008	1.012	1.016
$[8, 14]$ GeV	0.640	0.820	0.916	0.986	0.989	0.993
$[14, 20]$ GeV	0.633	0.810	0.906	0.977	0.979	0.981
$[20, 28]$ GeV	0.638	0.813	0.903	0.971	0.972	0.974
$[28, 40]$ GeV	0.614	0.794	0.887	0.959	0.960	0.961

Table 2: Mean containment $\langle f_{\text{contain}}(R)\rangle$, post-common, by truth photon p_T bin. Small R under-contains the photon; higher p_T marginally broadens the shower profile so containment decreases with p_T at fixed R.

Observations. Containment reaches $\sim$95% at $R = 0.10$ and saturates above unity by $R = 0.20$ (the slight excess over 1.0 reflects residual background plus cluster-shape calibration details). The pT dependence is mild: at $R = 0.10$ the average contained fraction drops from 92% in $[8, 14]$ GeV to 89% in $[28, 40]$ GeV, a $\sim$3 percentage-point broadening consistent with increased lateral spread at higher energies.

10 Implication for the inner-R scan

At $R_{\text{in}} = 0.05$ the inner ring removes only $\sim$66% of the photon's energy while the outer R = 0.4 cone contains essentially all of it; subtracting the former from the latter therefore leaves $\approx$ 34% of the photon in the recoiso$_{ET}$ definition. That $\sim$4 GeV contamination at $p_T^\gamma \sim 10$ GeV shifts the observed `cluster_iso_topo_04` − `cluster_iso_topo_005` mean from near zero to +3.9 GeV, which is what was seen in the inner-R scan of the purity MC closure and explains why the ABCD iso

region collapses at that working point. Inner radii of $R_{\text{in}} \geq 0.10$ leave $\lesssim 5\%$ photon residue and are the physics-motivated range for an annulus-isolation study if a clean photon-independent isolation variable is desired.

11 Files

- Analysis: `efficiencytool/iso_topo_containment_study.py`, `efficiencytool/iso_topo_make_rootfile.`
- Histograms: `efficiencytool/results/iso_topo_containment.root`.
- Arrays: `efficiencytool/results/iso_topo_containment.npz`.
- Topo plot macros: `plotting/plot_iso_topo_containment.C`, `plot_iso_topo_per_clusterEt.C`.
- Tower plot macros: `plot_iso_tower_per_clusterEt.C` (plain), `plot_iso_tower_thresh_r03_per_clusterEt` `plot_iso_tower_emcal70_per_clusterEt.C`, `plot_iso_tower_sub1_per_clusterEt.C`, `plot_iso_tower_excl`
- Figures: `plotting/figures/iso_topo_{dist_overlay,dist_data_vs_mc,containment_vs_R,dist_overlay` `iso_tower_{dist_overlay_per_clEt,thresh_r03_per_clEt,emcal70_per_clEt,sub1_per_clEt,excl_per_c`